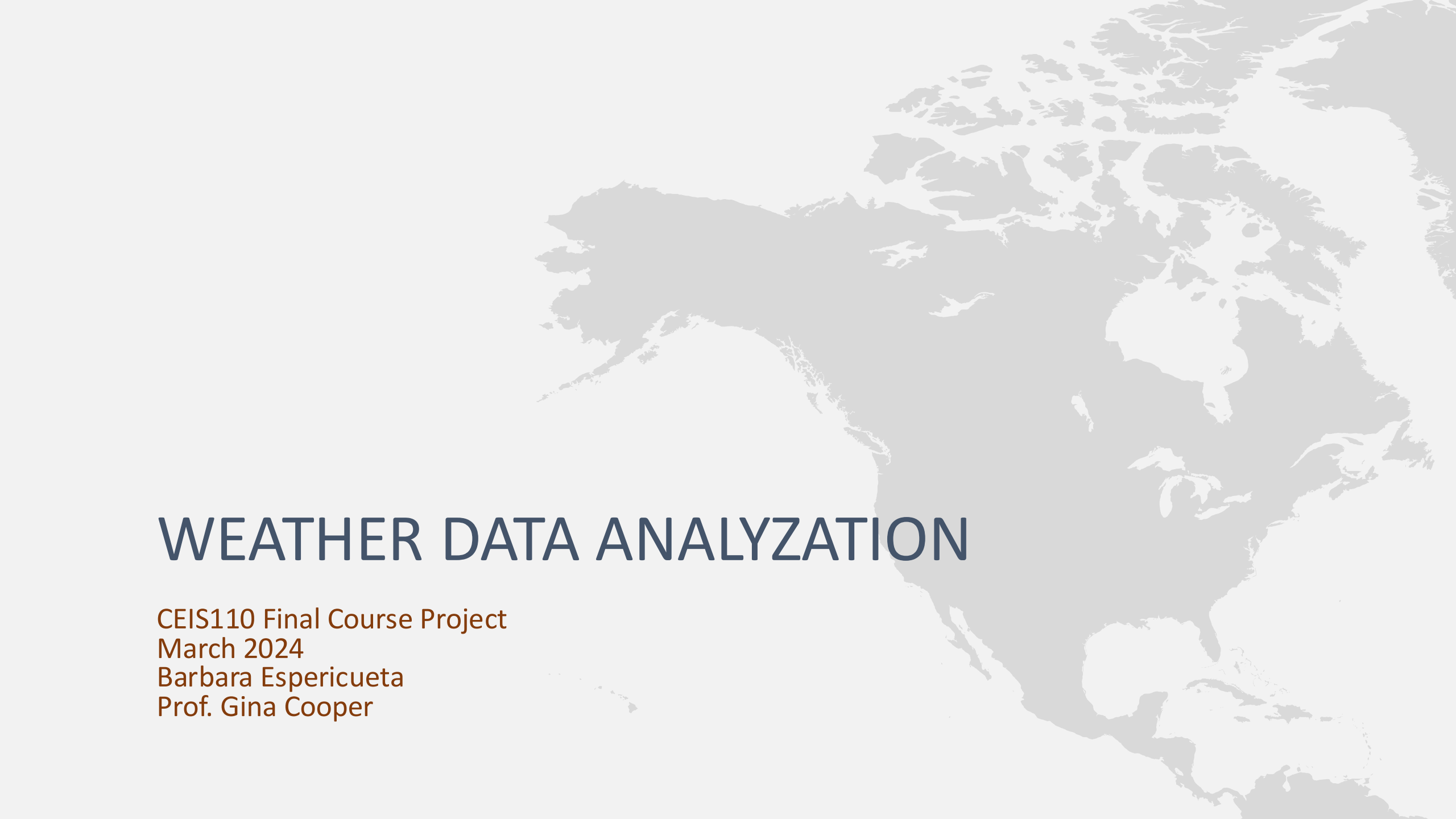




WEATHER DATA ANALYZATION

CEIS110 Final Course Project
March 2024
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Prof. Gina Cooper

INTRODUCTION

USING PYTHON TO PROGRAM DATA CLEANSING, ORGANIZATION, AND TRACKING.
DOWNLOADING A CLOUD-BASED APPLICATION PROGRAM INTERFACE (API) TO COLLECT DATA
FROM NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION (NOAA). USE STRUCTURED
QUERY LANGUAGE (SQL) AND PYTHON QUERYING THE WEATHER DATABASE(NOAA).

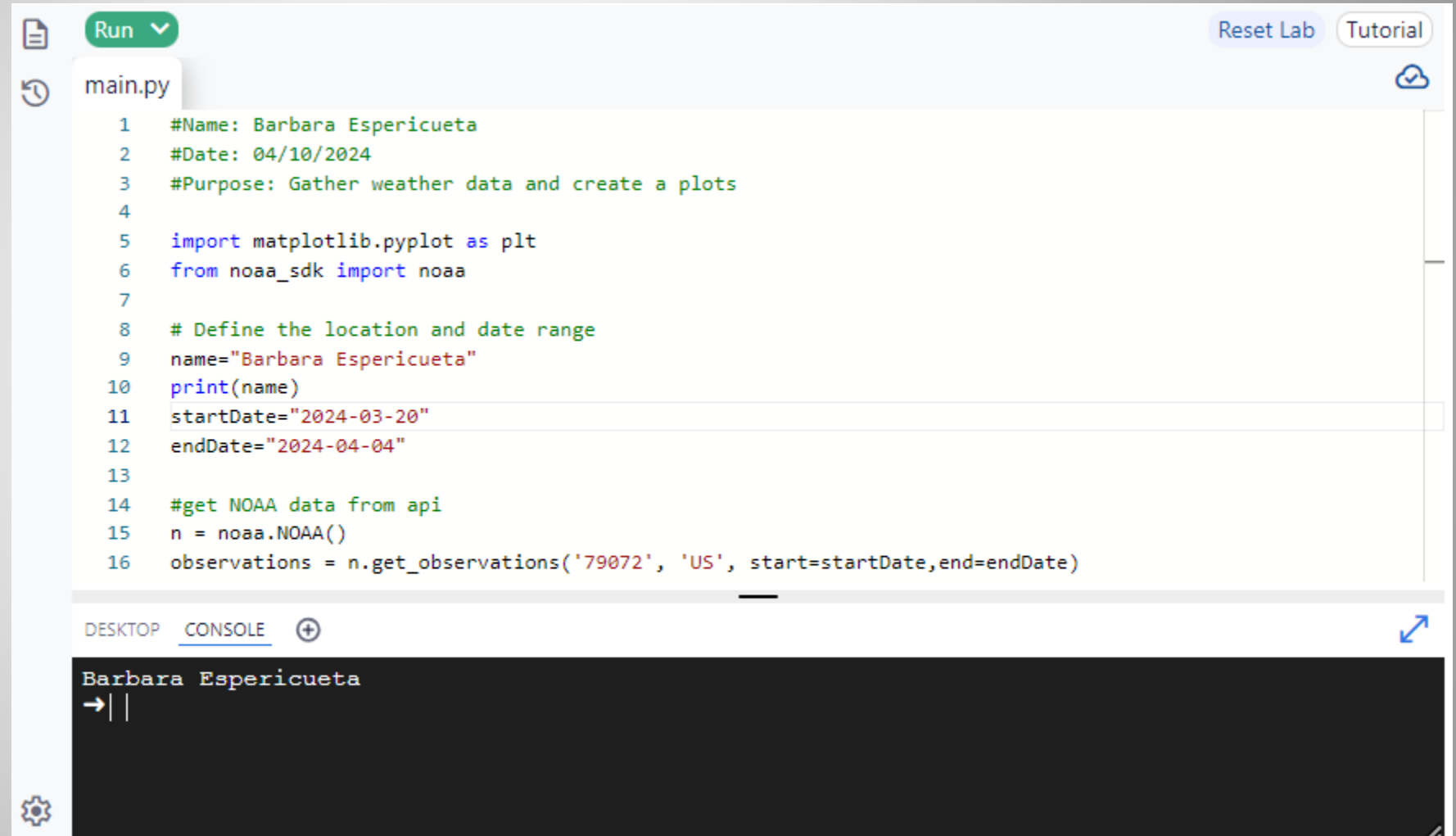
PROGRAM PREPARATION

Installing the NOAA database so we can access the weather data.

```
→ pip install noaa-sdk
Collecting noaa-sdk
  Downloading noaa_sdk-0.1.21-py3-none-any.whl.metadata (2.0 kB)
Collecting requests>=2.22.0 (from noaa-sdk)
  Downloading requests-2.31.0-py3-none-any.whl.metadata (4.6 kB)
Collecting charset-normalizer<4,>=2 (from requests>=2.22.0->noaa-sdk)
  Downloading charset_normalizer-3.3.2-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (33 kB)
Collecting idna<4,>=2.5 (from requests>=2.22.0->noaa-sdk)
  Downloading idna-3.7-py3-none-any.whl.metadata (9.9 kB)
Collecting urllib3<3,>=1.21.1 (from requests>=2.22.0->noaa-sdk)
  Downloading urllib3-2.2.1-py3-none-any.whl.metadata (6.4 kB)
Collecting certifi>=2017.4.17 (from requests>=2.22.0->noaa-sdk)
  Downloading certifi-2024.2.2-py3-none-any.whl.metadata (2.2 kB)
Downloading noaa_sdk-0.1.21-py3-none-any.whl (11 kB)
Downloading requests-2.31.0-py3-none-any.whl (62 kB)
----- 62.6/62.6 kB 9.5 MB/s eta 0:00:00
Downloading certifi-2024.2.2-py3-none-any.whl (163 kB)
----- 163.8/163.8 kB 15.4 MB/s eta 0:00:00
Downloading charset_normalizer-3.3.2-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (142 kB)
----- 142.1/142.1 kB 21.1 MB/s eta 0:00:00
Downloading idna-3.7-py3-none-any.whl (66 kB)
----- 66.8/66.8 kB 9.8 MB/s eta 0:00:00
Downloading urllib3-2.2.1-py3-none-any.whl (121 kB)
----- 121.1/121.1 kB 17.5 MB/s eta 0:00:00
Installing collected packages: urllib3, idna, charset-normalizer, certifi, requests, noaa-sdk
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
botocore 1.29.40 requires urllib3<1.27,>=1.25.4, but you have urllib3 2.2.1 which is incompatible.
Successfully installed certifi-2024.2.2 charset-normalizer-3.3.2 idna-3.7 noaa-sdk-0.1.21 requests-2.31.0 urllib3-2.2.1
```

IMPORTING LIBRARIES

Code to gather weather data for a specific number of days and then writing codes to import libraries from NOAA.



The screenshot displays a Jupyter Lab environment. At the top, there is a 'Run' button and a 'Tutorial' button. The main area shows a code editor with a file named 'main.py'. The code in the editor is as follows:

```
1 #Name: Barbara Espericueta
2 #Date: 04/10/2024
3 #Purpose: Gather weather data and create a plots
4
5 import matplotlib.pyplot as plt
6 from noaa_sdk import noaa
7
8 # Define the location and date range
9 name="Barbara Espericueta"
10 print(name)
11 startDate="2024-03-20"
12 endDate="2024-04-04"
13
14 #get NOAA data from api
15 n = noaa.NOAA()
16 observations = n.get_observations('79072', 'US', start=startDate,end=endDate)
```

Below the code editor, there is a 'DESKTOP' and 'CONSOLE' tab. The 'CONSOLE' tab is active, showing the output of the code:

```
Barbara Espericueta
→ | |
```

TWO LIST TO STORE TEMPERATURE AND HUMIDITY



Run



main.py

```
14 #get NOAA data from api
15 n = noaa.NOAA()
16 observations = n.get_observations('79072','US', start=startDate, end=endDate)
17
18 #create two list to store temperature and humidity
19 temp = []
20 humidity = []
21
22 for obs in observations:
23     temp.append(obs['temperature']['value'])
24     humidity.append(obs['relativeHumidity']['value'])
25     print(obs['timestamp'], "\t",
26           obs['temperature']['value'], "\t",
27           obs['relativeHumidity']['value'], "\t",
28           obs['textDescription'])
```

DESKTOP CONSOLE

```
Barbara Espericueta
2024-04-18T23:50:00+00:00      15.9      45.710445498964      Clear
2024-04-18T23:30:00+00:00      15.2      47.472857258479      Clear
2024-04-18T23:10:00+00:00      None      None      Clear
2024-04-18T22:50:00+00:00      16.2      45.476102368991      Clear
2024-04-18T22:30:00+00:00      16.3      45.504399334242      Clear
```

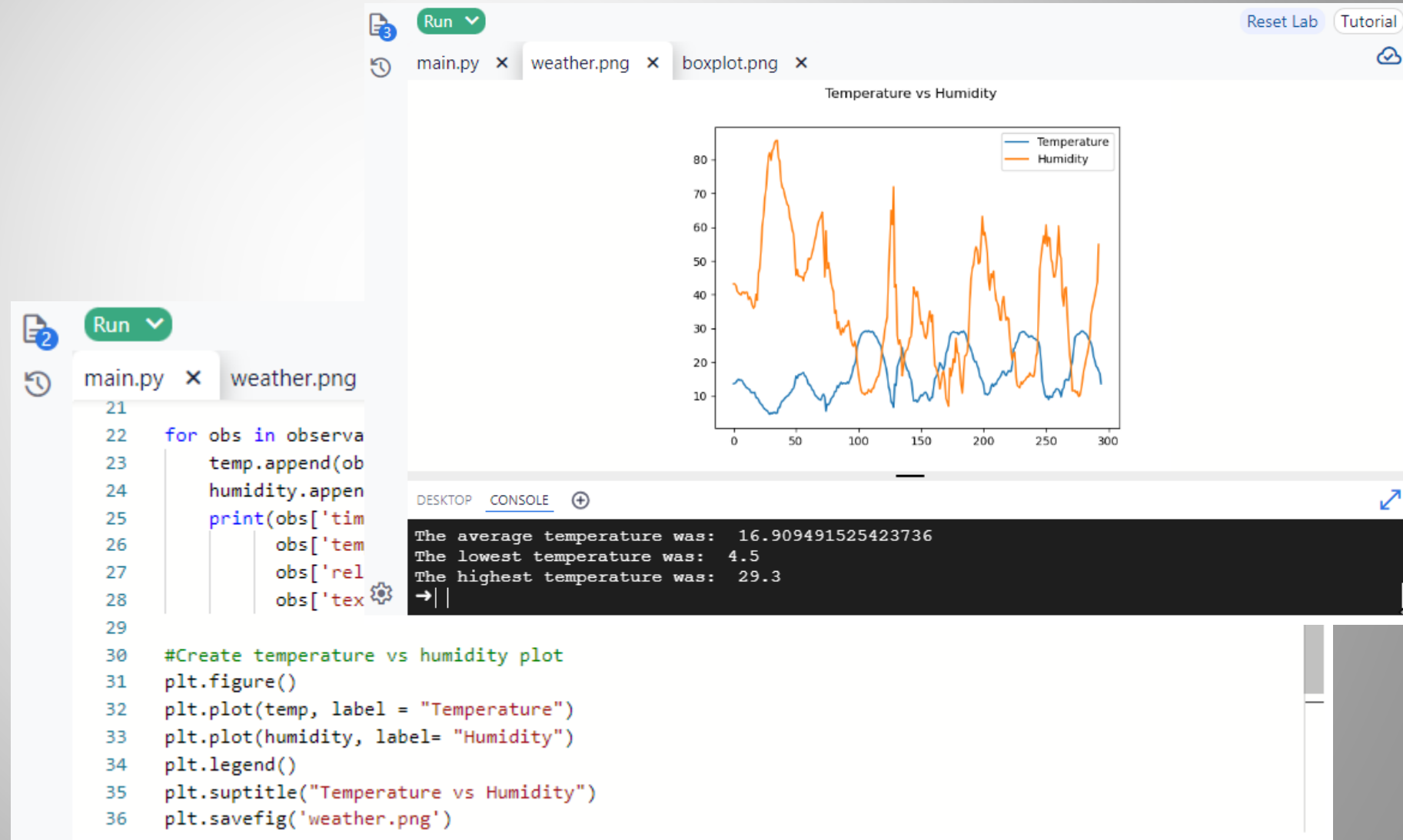
DESKTOP CONSOLE

```
Barbara Espericueta
2024-04-18T23:50:00+00:00      15.9      45.710445498964      Clear
2024-04-18T23:30:00+00:00      15.2      47.472857258479      Clear
2024-04-18T23:10:00+00:00      None      None      Clear
2024-04-18T22:50:00+00:00      16.2      45.476102368991      Clear
2024-04-18T22:30:00+00:00      16.3      45.504399334242      Clear
2024-04-18T22:10:00+00:00      None      None      Clear
2024-04-18T21:50:00+00:00      16.4      45.215616022527      Clear
2024-04-18T21:10:00+00:00      None      None      Clear
2024-04-18T20:50:00+00:00      16.7      45.30053920386      Clear
2024-04-18T20:10:00+00:00      16.9      44.109329656583      Clear and Windy
2024-04-18T19:50:00+00:00      15.9      46.355071978959      Clear
2024-04-18T19:30:00+00:00      None      None      Clear
2024-04-18T19:10:00+00:00      None      None      Clear
2024-04-18T18:50:00+00:00      15.7      46.624155670653      Clear and Windy
2024-04-18T18:30:00+00:00      None      None      Clear
2024-04-18T18:10:00+00:00      None      None      Clear
2024-04-18T17:50:00+00:00      14.6      47.638933785542      Clear and Windy
2024-04-18T17:30:00+00:00      13.4      50.772616049619      Clear
2024-04-18T17:10:00+00:00      13.3      50.388133878805      Clear and Windy
2024-04-18T16:50:00+00:00      12.7      50.940400948162      Clear and Windy
2024-04-18T16:30:00+00:00      11.8      52.531523343766      Clear and Breezy
2024-04-18T16:10:00+00:00      11.5      53.964812534008      Clear and Windy
2024-04-18T15:50:00+00:00      10.4      56.023774459975      Clear and Breezy
2024-04-18T15:30:00+00:00      9.9      57.515232763081      Clear and Breezy
```

Creating two list, a temperature list and a humidity list using a for loop.

TEMPERATURE VS HUMIDITY PLOT

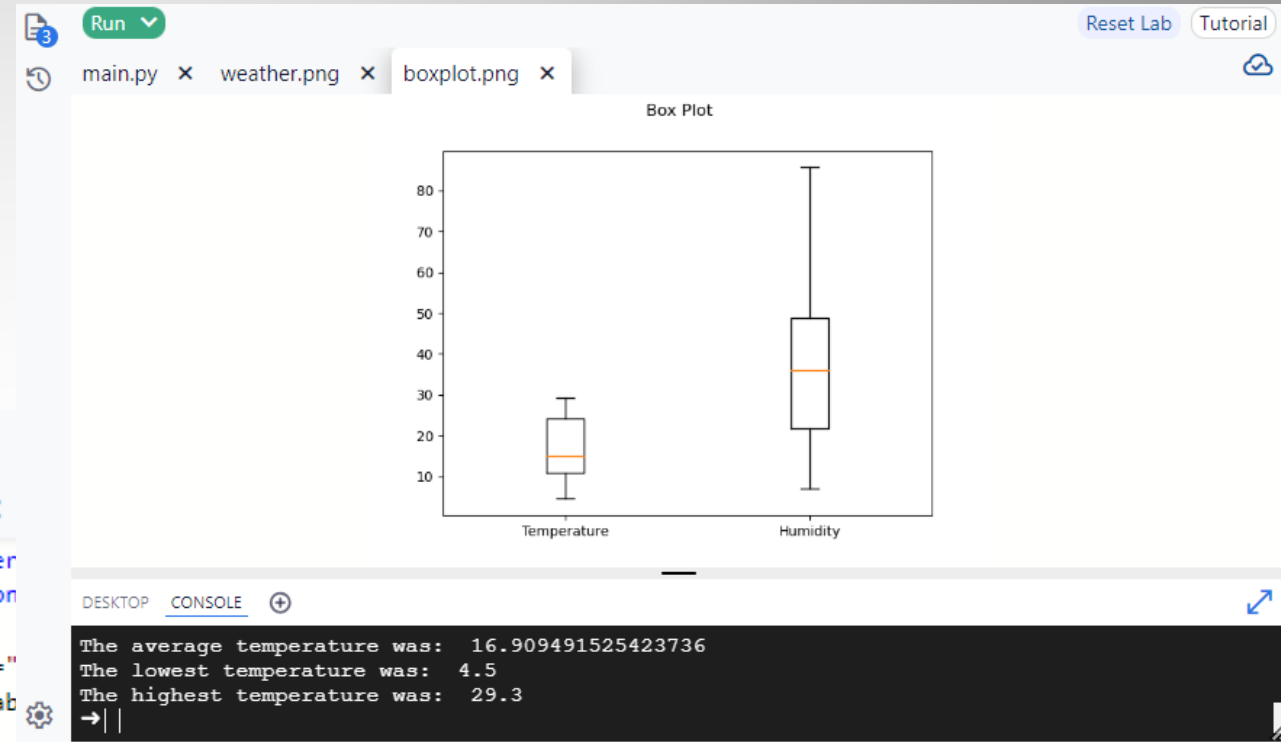
Creating a plot that shows temperature versus humidity in Texas.



BOX PLOT

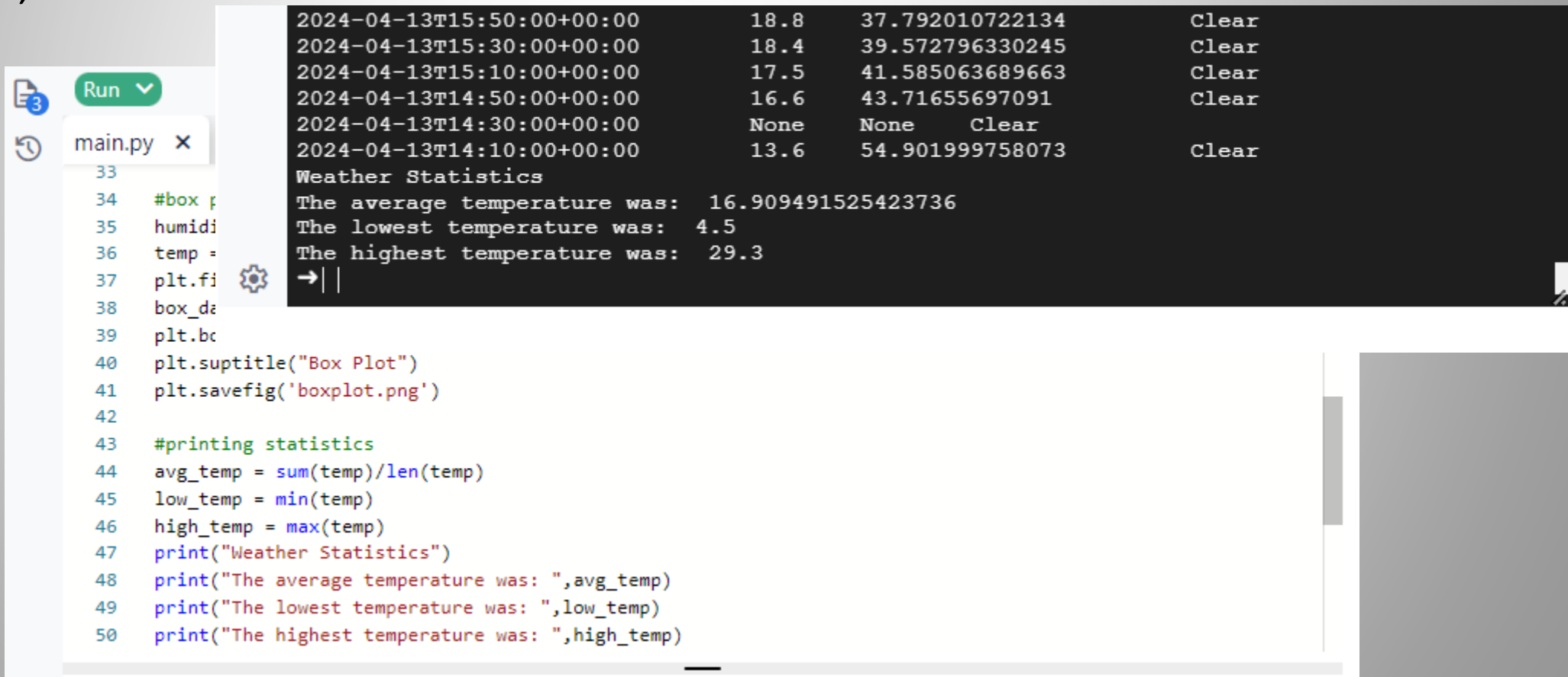
This box plot shows the statistical distribution of the data. The minimum, maximum, and median can be seen with the box plot

```
main.py x weather.png x
25 humidity = list(filter
26 temp = list(filter(Non
27 plt.figure()
28 plt.plot(temp, label="
29 plt.plot(humidity, lab
30 plt.legend()
31 plt.suptitle("Temperature vs Humidity")
32 plt.savefig('weather.png')
33
34 #box plot
35 humidity = list(filter(None, humidity))
36 temp = list(filter(None, temp))
37 plt.figure()
38 box_data = [temp, humidity]
39 plt.boxplot(box_data, labels = ['Temperature', 'Humidity'])
40 plt.suptitle("Box Plot")
41 plt.savefig('boxplot.png')
42
```



WEATHER STATISTICS

Printing the
average, lowest,
and highest
temperatures.



The image shows a Python IDE with a file named 'main.py'. The code in the editor calculates weather statistics and creates a box plot. The output window shows a table of weather data and the calculated statistics.

```
33
34 #box p
35 humidi
36 temp =
37 plt.fi
38 box_da
39 plt.bc
40 plt.suptitle("Box Plot")
41 plt.savefig('boxplot.png')
42
43 #printing statistics
44 avg_temp = sum(temp)/len(temp)
45 low_temp = min(temp)
46 high_temp = max(temp)
47 print("Weather Statistics")
48 print("The average temperature was: ",avg_temp)
49 print("The lowest temperature was: ",low_temp)
50 print("The highest temperature was: ",high_temp)
```

Timestamp	Temperature (°C)	Humidity (%)	Weather
2024-04-13T15:50:00+00:00	18.8	37.792010722134	Clear
2024-04-13T15:30:00+00:00	18.4	39.572796330245	Clear
2024-04-13T15:10:00+00:00	17.5	41.585063689663	Clear
2024-04-13T14:50:00+00:00	16.6	43.71655697091	Clear
2024-04-13T14:30:00+00:00	None	None	Clear
2024-04-13T14:10:00+00:00	13.6	54.901999758073	Clear

Weather Statistics
The average temperature was: 16.909491525423736
The lowest temperature was: 4.5
The highest temperature was: 29.3

CHALLENGES/LESSONS LEARNED

Challenges I faces during this project was organization of the program's language. There are places, an organization where the codes should be properly placed, more like the language of the program. Lessons learned was an understanding and respect for Python.

CAREER SKILLS

Skills I obtained during this project knowledge of installing and generating code from different sources to create anything needed. This can be done with literally anything. Organization and critical thinking skill were sharpened.

CONCLUSION

This project required an understanding of programming. Using Python to program language to manipulative data from sources into formats so we can easily analyze them in useful tactics. Even though there were challenges, there were more skills obtained during this project. The skills of critical thinking, patience, and organization a crucial in my career in Cybersecurity.